Bundler was primarily designed for use on a long-running system where installed gems hang around; the main use case is updating the set of *currently installed* gems. However, rebuilding an image is akin to building a new machine from scratch—in this scenario, Bundler is *not* being used on a long-running system. This, in essence, is the source of the issue.

Let's have another look at our Dockerfile and walk through what happens when we rebuild our image. Here are the last seven lines:

```
Line 1 COPY Gemfile* /usr/src/app/
  2 WORKDIR /usr/src/app
  3 RUN bundle install
  4
  5 COPY . /usr/src/app/
  6
  7 CMD ["bin/rails", "s", "-b", "0.0.0.0"]
```

When we modify our Gemfile, it busts the cache for line 1 of the extract. That means the cached intermediate layers for subsequent steps are thrown away, including the gems previously installed in the bundle install step (line 3). When the image build reaches this step again, it's as if we've never run bundle install on this machine before. No wonder the gems must be reinstalled from scratch.

How big a problem is this?

It depends on your situation. You may have more or less gems in your project(s), and more or less tolerance for waiting while gems install. The thing our basic approach has going for it is just that—it's basic. No extra things to configure or remember—just rebuild your image and you're done. This may be a perfectly acceptable option for many people.

However, if the time waiting for gems to build bothers you, then it's worth considering another option.

Using a Gem Cache Volume

As we've seen, the key problem is that Docker's image building, akin to building a machine from scratch, is at odds with the caching of gems with Bundler. What if, instead of fighting the image build process, we bypassed it? We've already seen on page 81 how volumes provide persistent file storage that's separate from the container filesystem; we can use this to solve our thorny problem.

Here's the gist.

By mounting a volume in the directory where Bundler installs our gems, we can execute Bundler commands to populate and manage the gems on this